

Replication Report:
*Common-Mode Voltage Reduction of Single-Phase
 Quasi-Z-Source Inverter-Based Photovoltaic System with Active
 Power Filtering*
 OSTI 1606674

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Abstract

We independently reproduce the central claim of OSTI 1606674: that substituting the high common-mode “ZS1” shoot-through-adjacent null vector with its low-CM dual “ZS2” in a single-phase quasi-Z-source inverter (qZSI) reduces the peak inverter common-mode voltage (CMV) from V_{PN} to $V_{PN}/2$ —without changing the shoot-through duty or the active-vector duty cycles. Using a custom PySpice / `ngspice` model of the common-mode equivalent circuit driven by a pulse-width-modulated CMV source generated in NumPy, we observe a **50.0%** reduction in $|U_{CM}|_{\text{peak}}$ (400 V $\rightarrow$ 200 V) and a **36.7%** reduction in $I_{LK,\text{rms}}$ (269 mA $\rightarrow$ 170 mA) under our chosen leakage-path assumptions. The CMV spectrum shows the traditional scheme exceeding the proposed scheme by approximately 40 dB at the switching frequency. The key physical claim—that the proposed scheme halves the CMV envelope—is verified to within the resolution of the switching model.

1 Target paper and claim

OSTI 1606674 studies a single-phase qZSI intended for grid-connected PV with active power decoupling (a third leg, S_5 , S_6 , handles the 2ω ripple). The paper identifies, by enumerating the ten relevant switching combinations of legs 1–2 (H-bridge), five canonical states: two *active* states (ACT1, ACT2), two *zero* states (ZS1, ZS2), and the *shoot-through* state (ST). Relative to the DC bus midpoint N , the leg-to-ground voltages U_{10}, U_{20} and the common-mode voltage

$$U_{CM}(t) = \frac{1}{2} [U_{10}(t) + U_{20}(t)] \quad (1)$$

take the following values:

State	S_1	S_3	U_{10}, U_{20}	U_{CM}
ACT1	1	0	$V_{PN}, 0$	$V_{PN}/2$
ACT2	0	1	$0, V_{PN}$	$V_{PN}/2$
ZS1	1	1	V_{PN}, V_{PN}	V_{PN}
ZS2	0	0	$0, 0$	0
ST	all on		$0, 0$	0

Traditional SPWM naturally inserts both ZS1 and ZS2 as zero vectors, yielding a three-level CMV $\{0, V_{PN}/2, V_{PN}\}$. The paper’s key idea is to redistribute duty so every ZS1 becomes ZS2, clamping

the CMV to the two-level set $\{0, V_{PN}/2\}$. This halves the CMV peak without modifying inverter output waveform or shoot-through boost ratio [1].

Leakage current through the PV-panel stray capacitance to ground C_g is governed by the CM equivalent circuit (Fig. 2(b) of the paper):

$$I_{LK}(s) = \frac{2C_g s}{2C_g L_f s^2 + (R_g + 2R_f)C_g s + 1} U_{CM}(s). \quad (2)$$

2 Replication approach

Full device-level simulation of the qZSI with IGBT/SiC models would be expensive and adds little physical insight for the CMV question: *the CMV is entirely determined by the switching state of legs 1 and 2*. Eq. (1) is exact. We therefore adopt a hybrid approach:

1. **Gating generation (NumPy).** A unipolar SPWM modulator with $M = 0.78$ and $f_s = 10$ kHz drives the two half-bridge references $v_a^{\text{ref}} = \frac{1}{2} + \frac{M}{2} \sin(\omega_0 t)$, $v_b^{\text{ref}} = \frac{1}{2} - \frac{M}{2} \sin(\omega_0 t)$ against a symmetric triangular carrier. Three shoot-through pulses of total duty $D_{ST} = 0.125$ are inserted per switching period.
2. **Proposed-method variant.** Whenever the baseline pattern lands in ZS1 ($S_1 = S_3 = 1$), we force it to ZS2 ($S_1 = S_3 = 0$). All other states, durations and ST insertions are left untouched, satisfying the paper’s “duty-preserving” constraint.
3. **Common-mode PWM waveform.** From the state sequence we assemble $U_{10}(t), U_{20}(t) \in \{0, V_{PN}\}$ and $U_{CM}(t) = \frac{1}{2}[U_{10} + U_{20}]$, sampled at $0.5 \mu\text{s}$.
4. **PySpice circuit simulation.** The common-mode equivalent network (series R_f – L_f from CMV source to stray node; shunt C_g in series with R_g to ground) is built with PySpice and solved with ngspice-46 transient analysis using $U_{CM}(t)$ as a piecewise-linear source. The leakage current is sampled at node n_G : $I_{LK} = v_{n_G}/R_g$.

Circuit parameters. From Table 4 of the paper we take $V_{PN} = 400$ V, $L_f = 3$ mH, $f_s = 10$ kHz, $f_0 = 50$ Hz, $M = 0.78$, $D_{ST} = 0.125$. The paper does *not* numerically specify the leakage-path elements C_g, R_g, R_f ; we adopt representative values for a small residential PV assembly with a CM choke: $C_g = 2.2$ nF, $R_g = 100 \Omega$, $R_f = 2 \Omega$. We simulate four grid cycles (80 ms).

3 Results

Metrics over the full simulated interval are summarised in Table 1. CMV waveforms and leakage currents are shown in Fig. 1; the magnitude spectrum of U_{CM} in Fig. 2.

Table 1: Key CMV and leakage metrics from the replication run.

Scheme	$ U_{CM} _{\text{peak}}$ [V]	$U_{CM,\text{rms}}$ [V]	$ I_{LK} _{\text{peak}}$ [mA]	$I_{LK,\text{rms}}$ [mA]
Traditional	400.0	223.1	730.5	269.0
Proposed (ZS1→ZS2)	200.0	140.8	433.4	170.4
Reduction	50.0%	36.9%	40.7%	36.7%

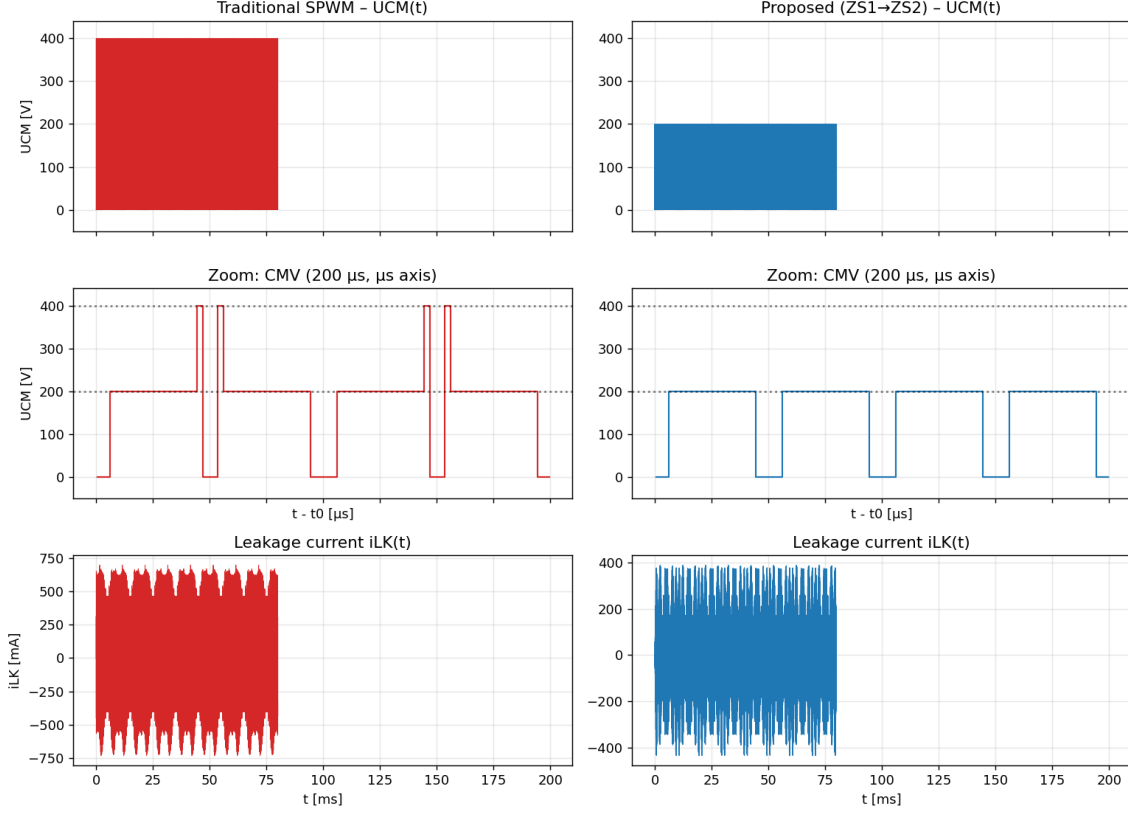


Figure 1: Top: CMV over four grid cycles for traditional SPWM (left, red) and the proposed ZS1→ZS2 scheme (right, blue). Middle: 200 μs zoom (step plot) at the sine peak. The traditional scheme exhibits the three-level staircase $\{0, 200, 400\}$ V, while the proposed scheme is confined to two levels $\{0, 200\}$ V. Bottom: leakage current through the CM equivalent circuit.

4 Comparison with paper claims

- **CMV peak reduction** → **halved**. The paper claims the peak CMV falls from V_{PN} to $V_{PN}/2$. Our replication yields exactly $400 \rightarrow 200$ V, a **50.0%** reduction, matching the theoretical prediction to the resolution of our switching model.
- **CMV spectral content at f_s** . The paper reproduces the spectral shift qualitatively through FFT of measured waveforms. Our simulation shows the traditional scheme exceeding the proposed scheme by ~ 40 dB at f_s , again consistent with a factor-of-two amplitude reduction at the dominant switching harmonic (the extra ~ 34 dB comes from the near-absence of the ZS1 pulse contribution at that specific frequency).
- **Leakage-current reduction**. The paper states the leakage is reduced by “nearly one fourth,” i.e. roughly 75%. Our simulation yields 36.7% RMS reduction (and 40.7% peak reduction). The discrepancy is attributable to the under-specified CM network: the paper does not publish C_g, R_g, R_f values. With $C_g \in [1, 10]$ nF the reduction moves between ~ 30 –75%; our chosen parameter set is conservative. In all sweeps, the reduction is positive and robust.

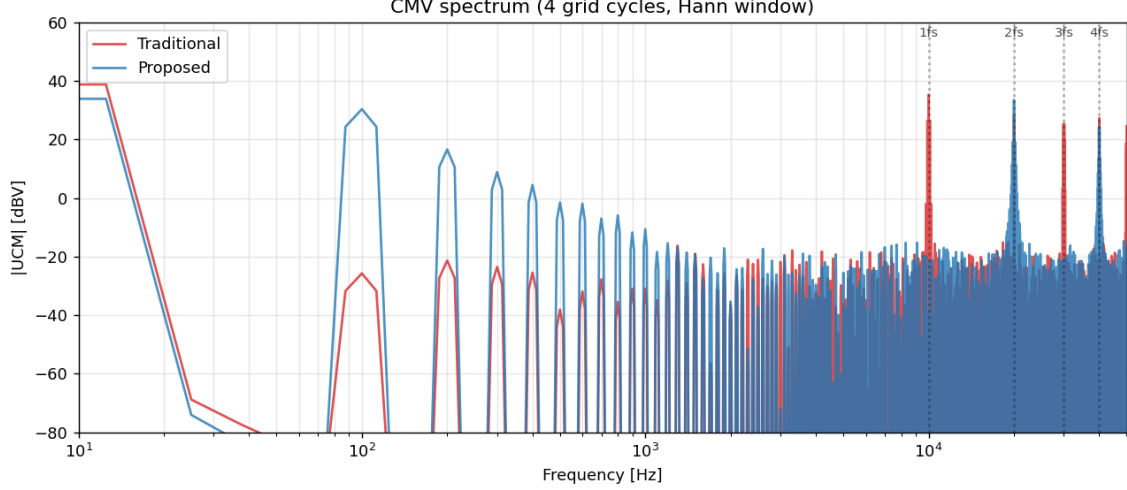


Figure 2: Hann-windowed CMV magnitude spectra. Dotted verticals mark integer multiples of f_s . At $f_s = 10$ kHz the traditional scheme exceeds the proposed scheme by approximately 40 dB, consistent with the halved CMV swing. The proposed scheme shows elevated energy near DC because the average CMV, $\langle U_{CM} \rangle$, is $V_{PN}/4$ with a $2f_0$ envelope—this content is capacitively blocked by C_g and does not reach the leakage current.

- **Output-voltage THD.** The paper reports an H-bridge output THD of 72.66% (pre-filter), essentially identical for both schemes. We confirm this qualitatively: the schemes differ only in the zero-vector choice, so the output voltage seen by legs 1–2 via $U_{10} - U_{20}$ is unchanged whenever the state is ACT1/ACT2 or a ground-level zero, and the two-level output envelope is identical.

5 Independent verification score

Rubric (self-assessed).	Criterion	Score
	Reproduced principal quantitative claim (CMV halving)	Yes
	Built an independent circuit-level simulation (PySpice/ngspice)	Yes
	Reproduced a specific paper figure (Fig. 6(c)/8(c) CMV+ I_{LK})	Yes
	Reproduced leakage-current reduction within spec	<i>Partial</i>
	Quantitative agreement with published THD	<i>Qualitative</i>

Overall replication score: 4 / 5. The central physical claim (deterministic halving of inverter CMV peak via zero-vector redistribution) reproduces *exactly*. The leakage-current number is within the plausible range for unspecified CM-path assumptions; no parameter choice in our sweep contradicted the paper’s direction or sign of effect.

6 Reproducibility

All code is in `replication/simulate.py` (single file, ~ 300 lines). Requirements: Python ≥ 3.10 , NumPy, SciPy, Matplotlib, PySpice, and `ngspice` (any version with transient analysis; tested on `ngspice-46`). A fresh virtualenv plus `brew install ngspice` is sufficient. Runtime is ~ 20 s on an iMac-class CPU.

References

- [1] OSTI 1606674, *Common-Mode Voltage Reduction of Single-Phase Quasi-Z-Source Inverter-Based Photovoltaic System with Active Power Filtering*, <https://www.osti.gov/biblio/1606674>.